

CHAPTER 1

INTRODUCTION

1.1 Project Overview

In recent years, the importance of health and wellness has increased significantly, especially after the global pandemic where individuals shifted towards home-based fitness activities. Yoga is one of the most effective practices for improving both physical and mental health. However, performing yoga without proper guidance often leads to incorrect posture, reduced effectiveness, and even injuries.

To address this issue, the Smart Yoga Mat Using IoT is developed as an intelligent system that assists users in performing yoga accurately without the need for constant supervision. The system integrates Internet of Things (IoT), computer vision, and sensor technologies to monitor posture, balance, and movement in real time.

The smart yoga mat uses a webcam and sensors to capture body posture and alignment. The captured data is processed using technologies like OpenCV and MediaPipe, which analyze body joints and angles. Based on this analysis, the system provides instant feedback to the user through a laptop or mobile application.

This project offers a low-cost, portable, and user-friendly solution that enhances home-based yoga practice by ensuring correct posture and improving overall performance.

1.2 Problem Statement

In home-based yoga practice, the absence of professional guidance leads to several challenges:

- Users often perform incorrect postures due to lack of supervision
- Improper body alignment increases the risk of injuries
- No real-time feedback system to correct mistakes instantly
- Lack of performance tracking reduces motivation and consistency
- Existing solutions depend heavily on cameras or lack sensor integration

Therefore, there is a need to develop a smart and intelligent system that can monitor posture, provide real-time feedback, and improve the effectiveness of yoga practice.

1.3 Objectives of Project

The main objectives of the Smart Yoga Mat Using IoT are:

- To design a **Smart Yoga Mat with an IoT-enabled image sensor** interfaced with a laptop for real-time posture analysis.
- To monitor **yoga session duration, pose accuracy, and type of yoga performed** in real time.
- To provide **live corrective feedback** and detailed performance tracking through the application.
- To promote **consistent yoga practice**, improve accuracy, and support overall health improvement using smart technology.

1.4 Scope Of Project

The Scope of the Smart Yoga Mat Using IOT are:

The Smart Yoga Mat is designed to enhance home-based yoga practice by integrating advanced pressure and motion sensors that monitor postures in real time. Connected to a mobile application, it provides immediate feedback on pose accuracy, weight distribution, and overall performance, ensuring practitioners maintain correct form without constant instructor supervision. The system tracks session duration, calorie expenditure, and live updates on pose accuracy while storing session history and analytics for long-term progress monitoring. Currently optimized for common yoga poses, it ensures safety and effectiveness, while its future scope includes expansion to support pilates, physiotherapy, and rehabilitation exercises with AI-driven personalized recommendation

CHAPTER 2

LITERATURE REVIEW

2.1 Review of literature available

Innovations in smart fitness systems require a thorough review of existing research work. Various researchers have proposed different methods for yoga posture detection using computer vision, IoT, and machine learning techniques. The relevant literature is discussed below:

[1] Long J., Jo K. H. (2019)

Developed a yoga posture coaching system using transfer learning. The system used a MobileNet-based model to recognize yoga poses with an accuracy of approximately 98.4%. It provided visual feedback for posture correction but relied only on camera-based input and did not include IoT sensor integration.

[2] Sun W. (2019)

Proposed RFitness, a smart yoga mat system using passive RFID technology. This approach enabled posture detection without cameras, thereby improving privacy. However, the system had limitations in detection accuracy due to RFID constraints and lacked real-time mobile application support.

[3] Gadhvi R., Desai P., Siddharth (2021)

Introduced PosePilot, an edge-AI-based posture correction system using LSTM models. The system provided real-time personalized feedback with low latency on mobile devices. However, it mainly relied on vision-based techniques and did not incorporate sensor-based data.

[4] Sharma A., Sharma P., Pincha D., Jain P. (2020)

Developed a computer vision-based yoga pose recognition system using MediaPipe and CNN models. The system achieved high accuracy in detecting yoga poses such as Surya Namaskar. Its performance was affected by lighting conditions and camera positioning.

[5] Anusha M., Dubey S., Raju P. S., Pasha I. A. (2021)

Proposed an embedded smart yoga mat using force-sensitive resistors (FSRs) to detect posture and balance. The system effectively measured pressure distribution but supported limited yoga poses and lacked full-body posture analysis.

CHAPTER 3

METHODOLOGY /SYSTEM DEVELOPMENT

3.1 Software Requirement Specification (SRS)

The Software Requirement Specification (SRS) describes the functional and non-functional requirements of the Smart Yoga Mat system. It defines how the system will operate, the resources required, and the interactions between hardware and software components. This document helps in understanding the complete system before implementation.

3.1.1 Software Requirements

The software tools required for the development of the Smart Yoga Mat system are:

- **Python** – Used as the main programming language for system development
- **OpenCV** – Used for image processing and frame analysis
- **MediaPipe** – Used for detecting body landmarks and joints
- **TensorFlow / PyTorch** – Used for implementing machine learning models
- **Android Studio / Flutter** – Used for mobile application development
- **Database (Firebase / SQLite)** – Used for storing user data and session history

3.1.2 Hardware Requirements

- **Smart Yoga Mat** – Base surface for performing yoga
- **Image Sensor (Webcam)** – Captures user posture
- **Laptop / Computer System** – Processes data and provides feedback
- **Microcontroller** – Main control unit used in the circuit to interface components

-
- **Ultrasonic Sensor** – Measures body alignment
 - **USB Interface** – Connects system components
 - **Power Supply** – Provides power to the circuit

3.1.3 Functional Requirements

The Smart Yoga Mat system is designed to perform the following functions:

- **Posture Capture:** The system captures real-time video of the user performing yoga using a webcam.
- **Body Landmark Detection:** The system identifies key body joints such as shoulders, elbows, hips, and knees using computer vision techniques.
- **Posture Analysis:** The system processes detected landmarks and calculates joint angles to analyze the user's posture.
- **Posture Comparison:** The system compares the user's posture with predefined correct yoga poses stored in the system.
- **Feedback Generation:** The system provides real-time feedback in the form of visual alerts or messages for posture correction.
- **Session Monitoring:** The system tracks session duration, type of pose, and overall performance during yoga practice.
- **Data Storage:** The system stores user performance data and session history for future analysis.
- **User Control:** The system allows the user to start, stop, and manage yoga sessions easily.

3.1.4 Algorithm / Technology / Mathematical Model

Algorithm (Step-by-Step Working)

- **Step 1:** Start the system and initialize components.
- **Step 2:** Capture real-time video using a webcam.
- **Step 3:** Convert the video into individual frames.
- **Step 4:** Apply preprocessing using OpenCV (resize, noise reduction).
- **Step 5:** Detect body landmarks using MediaPipe.
- **Step 6:** Calculate joint angles from detected landmarks.
- **Step 7:** Compare calculated angles with predefined reference poses.
- **Step 8:** If posture is correct, display a success message.
- **Step 9:** If posture is incorrect, provide correction feedback.
- **Step 10:** Store session data and performance details.
- **Step 11:** End the process.

Technology Used

- **Computer Vision:** Used for image processing and posture detection.
- **Machine Learning:** Used for analyzing and recognizing yoga poses.
- **IoT (Internet of Things):** Used for smart monitoring and data tracking.

Mathematical Model

Joint Angle Calculation

The system calculates joint angles using three body points **A, B, and C**, where the angle is measured at **point B (middle joint)**.

$$\theta = \cos^{-1} \left(\frac{AB^2 + BC^2 - AC^2}{2 \cdot AB \cdot BC} \right)$$

Meaning of Points

- **A** → First joint (e.g., Shoulder)
- **B** → Middle joint (e.g., Elbow) (*angle is calculated here*)
- **C** → Third joint (e.g., Wrist)

Distance Formula

Distances between points are calculated as:

$$AB = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$BC = \sqrt{(x_3 - x_2)^2 + (y_3 - y_2)^2}$$

$$AC = \sqrt{(x_3 - x_1)^2 + (y_3 - y_1)^2}$$

Example

For elbow angle:

A = Shoulder, B = Elbow, C = Wrist

3.2 PROJECT PLAN

The project plan defines the overall strategy and steps required to successfully complete the Smart Yoga Mat using IoT. It includes scheduling of activities, development process, feasibility analysis, risk management, and cost estimation. A proper project plan helps in organizing tasks, managing time efficiently, and ensuring the project is completed within the given timeline. It also helps in identifying potential risks and planning solutions in advance.

3.2.1 Project Schedule

Phase	Activity	Duration
Phase 1	Problem Identification & Topic Selection	1 week
Phase 2	Literature Review & Research	2 week
Phase 3	Requirement Analysis	1 week
Phase 4	System Design (Architecture, Diagrams)	2 week
Phase 5	Implementation (Coding & Integration)	4 week
Phase 6	Testing & Debugging	2 week
Phase 7	Documentation & Report Writing	2 week

Table 3.2.1: Project Schedule

3.2.2 SDLC (Software Development Life Cycle)

The project follows the Waterfall Model of SDLC:

1. Requirement Analysis – Understanding system needs like posture detection, feedback system, and tracking
2. System Design – Designing architecture, block diagram, and modules
3. Implementation – Developing code using Python, OpenCV, MediaPipe
4. Testing – Checking accuracy of posture detection and fixing errors
5. Deployment – Running the system for users
6. Maintenance – Improving performance and fixing future issues

3.2.3 Feasibility Study

Feasibility study is carried out to determine whether the proposed system is practical and achievable. It evaluates the project in terms of technical, economic, and operational aspects to ensure successful implementation.

1. Technical Feasibility

The proposed Smart Yoga Mat system is technically feasible as it uses widely available technologies such as Python, OpenCV, MediaPipe, and basic IoT sensors. The required hardware like webcam and sensors are easily accessible and compatible with the system. Hence, the development and implementation of the system are technically possible.

2. Economic Feasibility

The project is economically feasible as it requires low investment. Most components such as laptop and software tools are already available, and additional hardware like sensors and modules are affordable. The estimated cost of the system is within ₹8800 to ₹15000, making it cost-effective.

3. Operational Feasibility

The system is user-friendly and easy to operate. It provides simple interfaces and real-time feedback, making it suitable for users at home without professional training. Therefore, the system is operationally feasible and can be easily adopted by users.

3.2.4 Risk Analysis & Planning

Risk analysis is the process of identifying potential problems that may occur during the development and operation of the project. Proper planning is required to minimize these risks and ensure smooth functioning of the system.

Identified Risks and Mitigation Plan

Risk	Impact	Mitigation Plan
Incorrect posture detection	Reduces system accuracy	Improve algorithm and use better training data
Lighting conditions affecting camera	Poor image processing	Use proper lighting setup and adjust camera position
Sensor inaccuracies	Wrong posture feedback	Calibrate sensors properly
Hardware failure	System may stop working	Use reliable components and maintain backup
User misuse or improper setup	Incorrect results	Provide clear instructions to users

Table 3.2.4: Risk Analysis & Planning

3.2.5 Project Cost Estimation

The approximate expenditure required for the development of the Smart Yoga Mat using IoT is shown below

Sr. No	Component / Item	Quantity	Minimum Cost (₹)	Ideal Cost (₹)
1	Sensor Module (Camera Module)	1	5000	6000
2	Laptop (Already Owned)	-	300	700
3	Connectivity Interface (USB Cable)	1	1000	2500
4	Protective Mat Layer (Non-slip, Durable)	1	1000	2000
5	Feedback System (Screen Alerts & Voice Guidance – Maintenance Cost)	-	300	500
6	Optional IMU Sensor(e.g MPU-6050)	1	500	1500
7	Power Supply (USB Powered / Rechargeable Battery)	-	0	1000
8	Miscellaneous (Wiring, Adhesives, Casing, Mounts)	-	1000	1300
Total			8800	15000

Table 3.2.5: Project Cost Estimation

3.3 SOFTWARE DESIGN

Software design describes how the Smart Yoga Mat system is developed to process data, analyze posture, and provide feedback. It defines the structure, components, and working of the software used in the system. The system uses computer vision and IoT technologies to detect posture and guide the user.

3.3.1 Data Flow Diagram (DFD)

The Data Flow Diagram (DFD) represents how data flows through the Smart Yoga Mat system. It shows how input data from the user is processed and how output is generated in the form of feedback. The system uses a laptop camera and Ultrasonic Sensor (HC-SR04) to collect data for posture analysis.

The data flow of the system is represented using Level 0 and Level 1 DFD.

1.Level 0 DFD

The Level 0 DFD shows the overall working of the system as a single process. The user performs yoga, and the system captures posture using the laptop camera and distance data using the Ultrasonic Sensor (HC-SR04). The system processes this data and provides feedback to the user in the form of screen alerts and voice guidance.

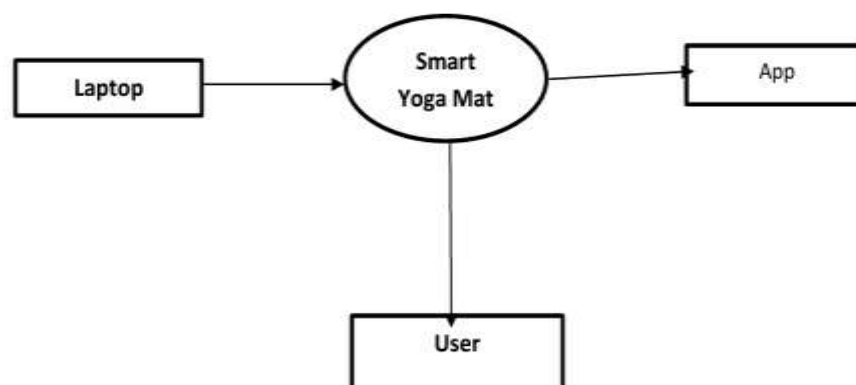


Figure 3.3.1.1Level 0 DFD

2.Level 1 DFD

The Level 1 DFD shows the detailed flow of data in the system. The laptop camera captures posture data, and the Ultrasonic Sensor (HC-SR04) measures distance and sends data through the microcontroller. The system processes this data using Python and OpenCV, analyzes posture, and provides feedback to the user through screen alerts and voice guidance.

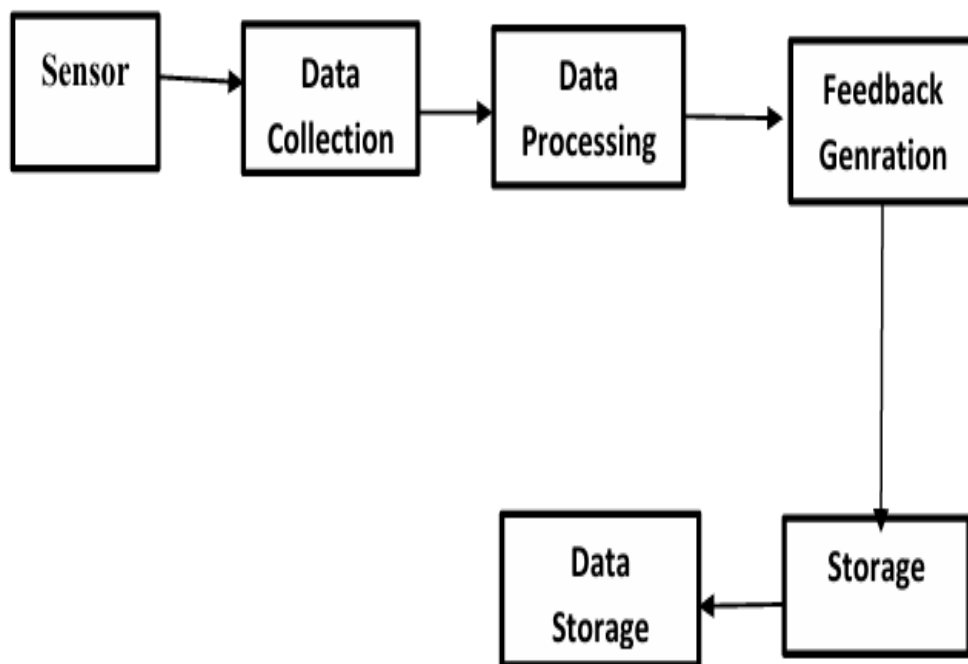


Figure 3.1.1.2 First level DFD

3.3.2 Flow Chart

The flowchart represents the step-by-step working of the Smart Yoga Mat system. It shows how the system captures input from the user, processes the data, and provides feedback based on posture analysis.

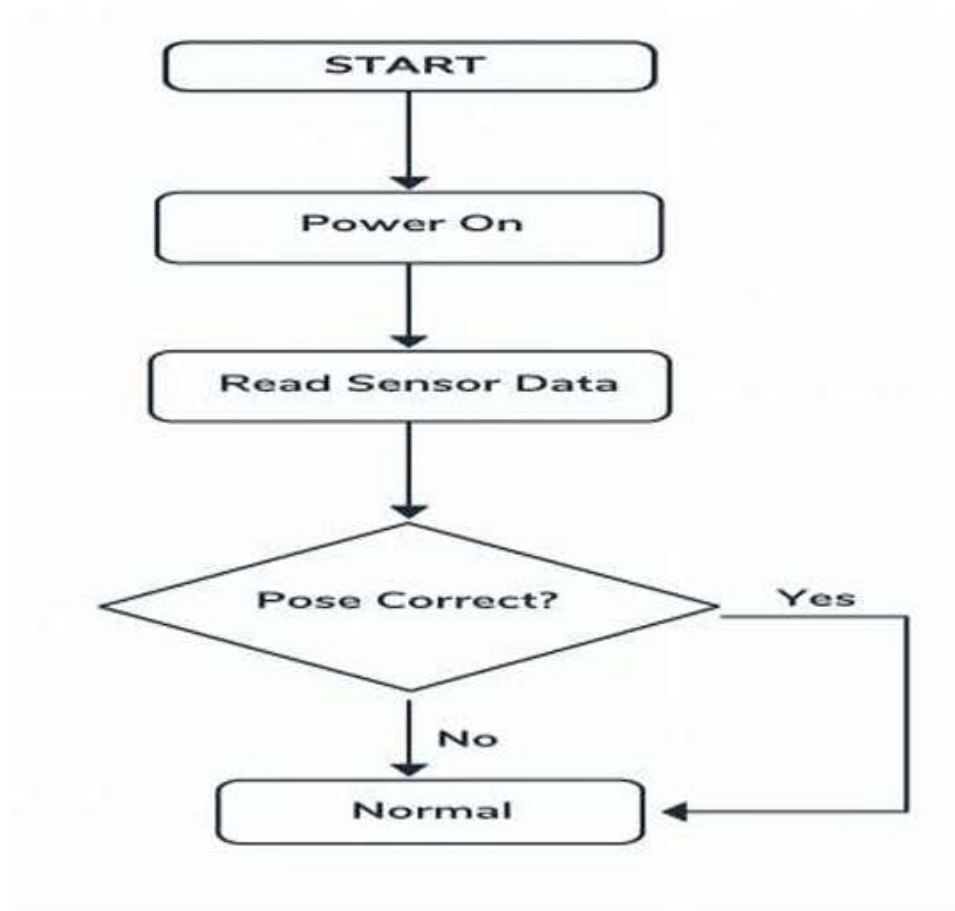


Figure 3.3.2. Flow chart

3.3.3 System Architecture

The system architecture of the Smart Yoga Mat using IoT represents the overall structure and interaction between hardware and software components. The system integrates a laptop camera, Ultrasonic Sensor (HC-SR04), and a microcontroller to capture and process data for posture detection.

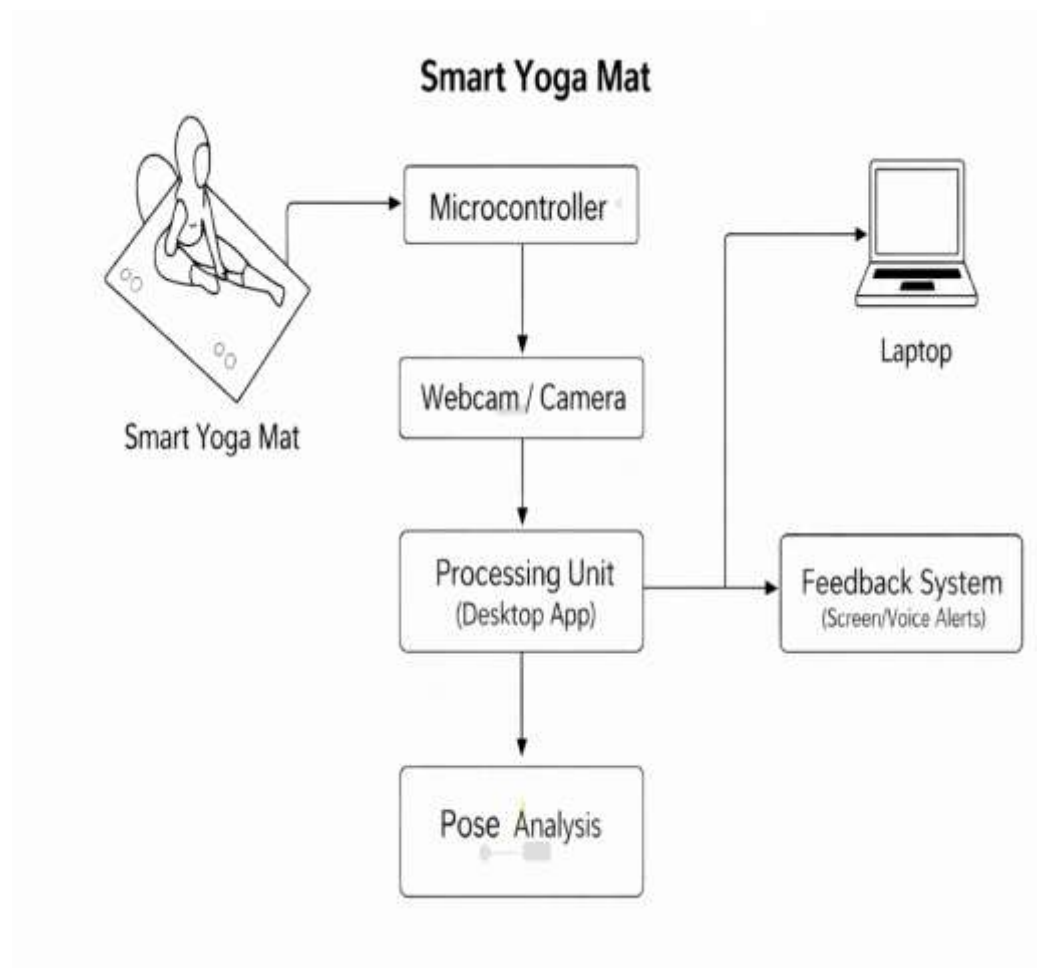


Figure 3.3.3 System Architecture Diagram

3.3.4 UML Diagrams

Unified Modeling Language (UML) is a standard technique used to design and visualize the structure and behavior of a system. It helps in understanding how different components interact and how the system works.

In the Smart Yoga Mat system, UML diagrams represent both structure and working of the system.

Types of UML Diagrams

1. Structural Diagrams

Show system structure and components.

Examples: Class, Component, Deployment

2. Behavioral Diagrams

Show system behavior and flow.

Examples: Use Case, Sequence, Activity, State Chart

1 Class Diagram

The Class Diagram defines the main classes of the Smart Yoga Mat system—such as User, Sensor, Processor, and Data Storage—and shows their attributes, methods, and relationships. It provides a structural blueprint of how the system is organized.

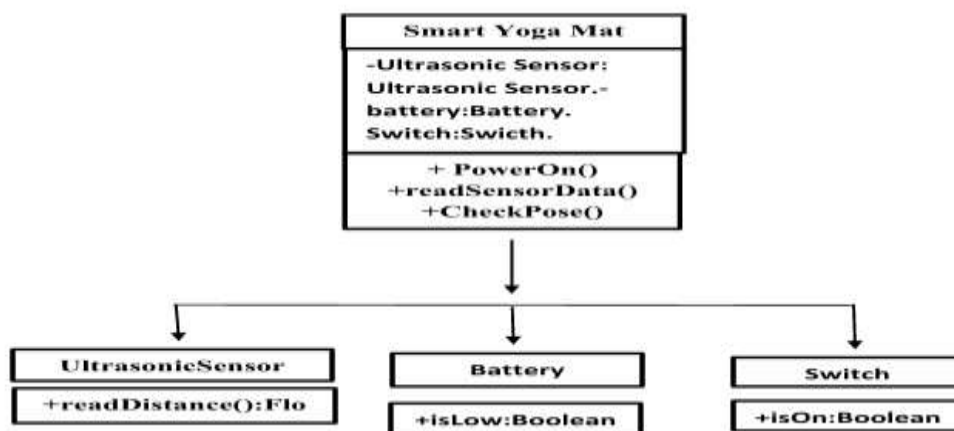


Fig 3.3.4.1 Class Diagram

2 Component Diagram

The Component Diagram represents different modules of the system and how they interact to perform overall functionality.

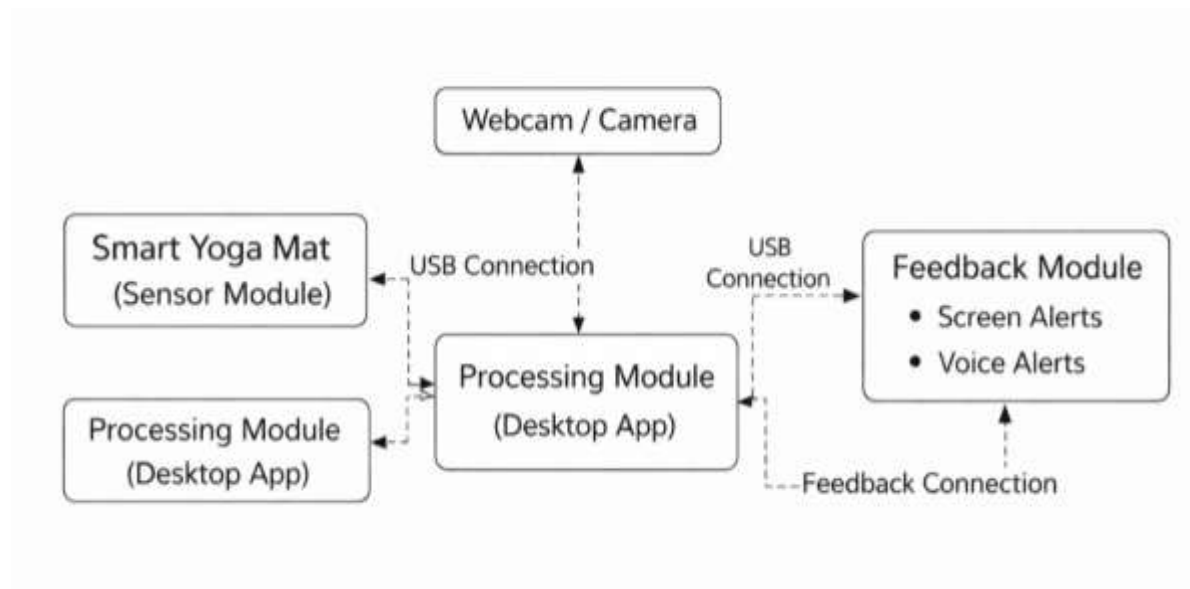
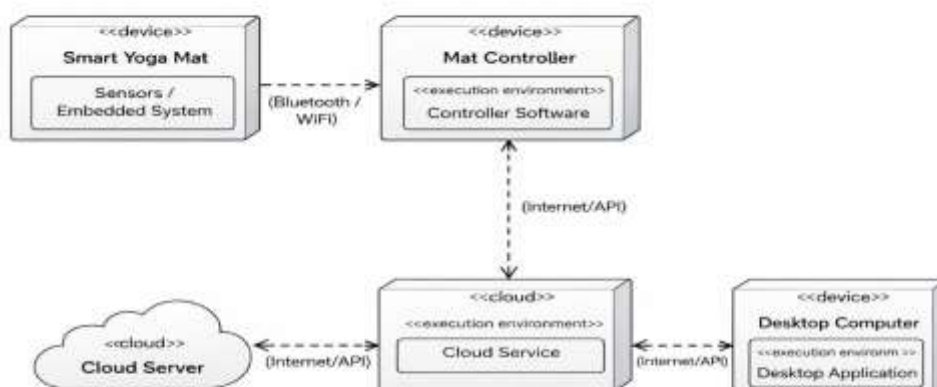


Fig 3.3.4.2 Component Diagram

3 Deployment Diagram

The Deployment Diagram depicts the system's hardware and software arrangement. It shows how components are deployed on nodes and how they communicate at runtime.



Fig

Figure 3.3.4.3 Deployment Diagram

4 Use Case Diagram

The Use Case Diagram illustrates how the user interacts with the Smart Yoga Mat system, highlighting key actions such as providing input, receiving feedback, and tracking yoga performance. It shows the functional requirements from the user's perspective.

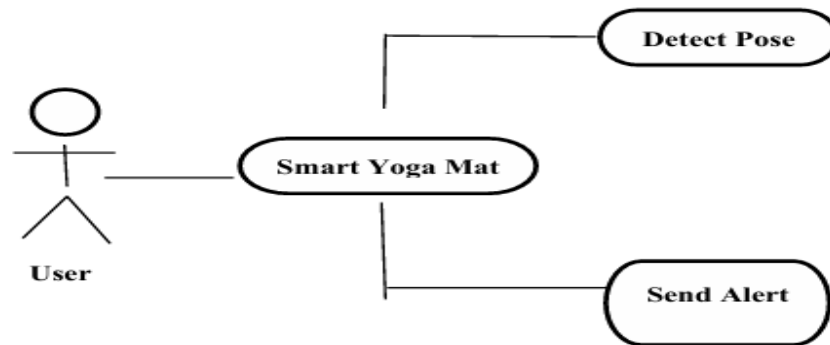


Figure 3.3.4.4: Use Case Diagram

5 Sequence Diagram

The Sequence Diagram shows the step-by-step interaction between the user, sensor module, processing module, and data storage, demonstrating how data flows in sequence to generate yoga feedback.

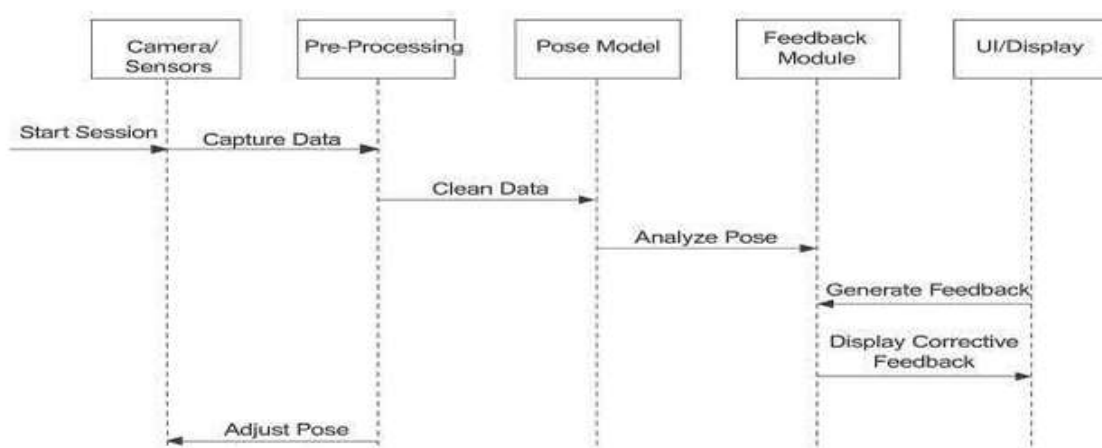


Figure 3.3.4.5: Sequence Diagram

6 Activity Diagram

This diagram shows the workflow of the system from starting the session to detecting posture and providing feedback.

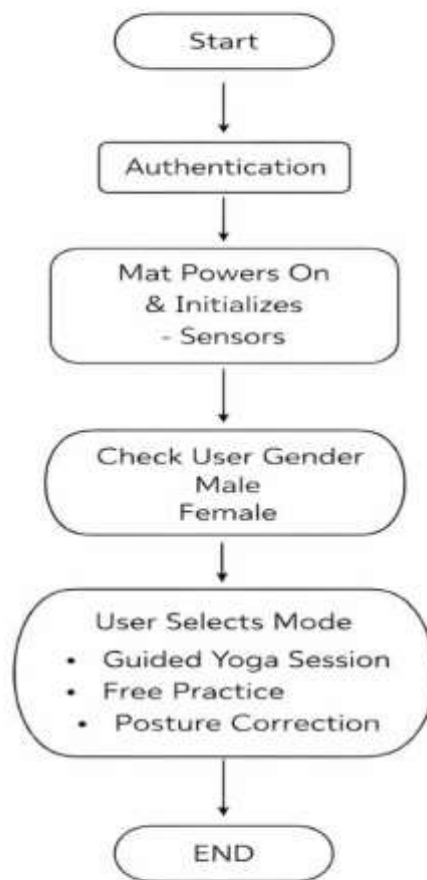


Figure 3.3.4.6: Activity Diagram

7 State Chart Diagram

The State Chart Diagram represents different states of the system such as idle, active, paused, and completed session.

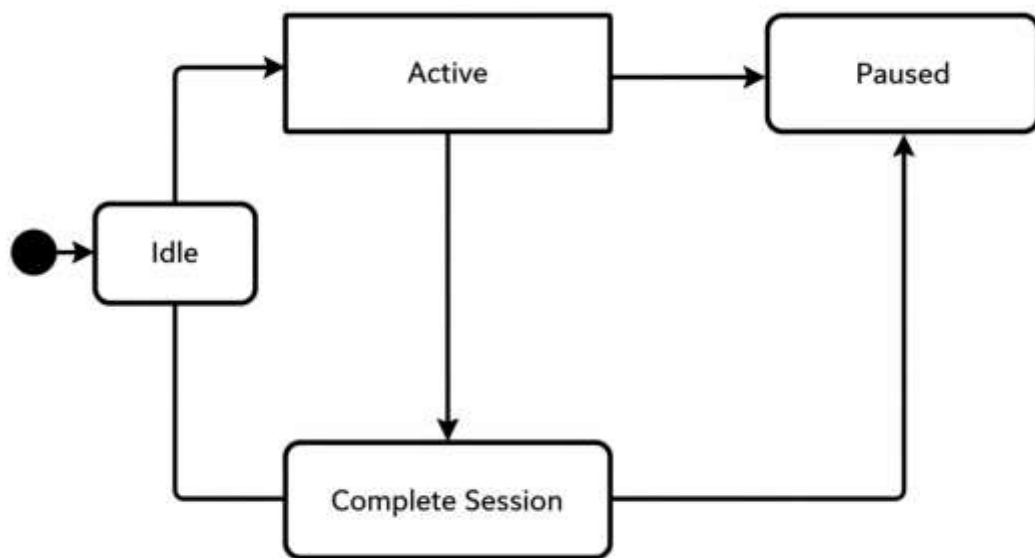


Figure 3.3.4.7: State Chart Diagram

3.3.5 Database Design / ER Diagram

The database design of the Smart Yoga Mat system stores and manages user information, yoga session details, pose data, and feedback. It is represented using an ER diagram that shows entities, attributes, and relationships. The main entities are User, Yoga Session, Pose Data, and Feedback Log. The User stores basic details, the Yoga Session records session information, Pose Data contains posture accuracy, and Feedback Log stores corrective messages.

These entities are connected through relationships, where a user performs multiple sessions, each session generates pose data, and feedback is provided accordingly. Primary and foreign keys ensure data consistency. This design supports efficient data storage and enables real-time posture analysis and feedback.

- **ER Diagram of Smart Yoga Mat System**

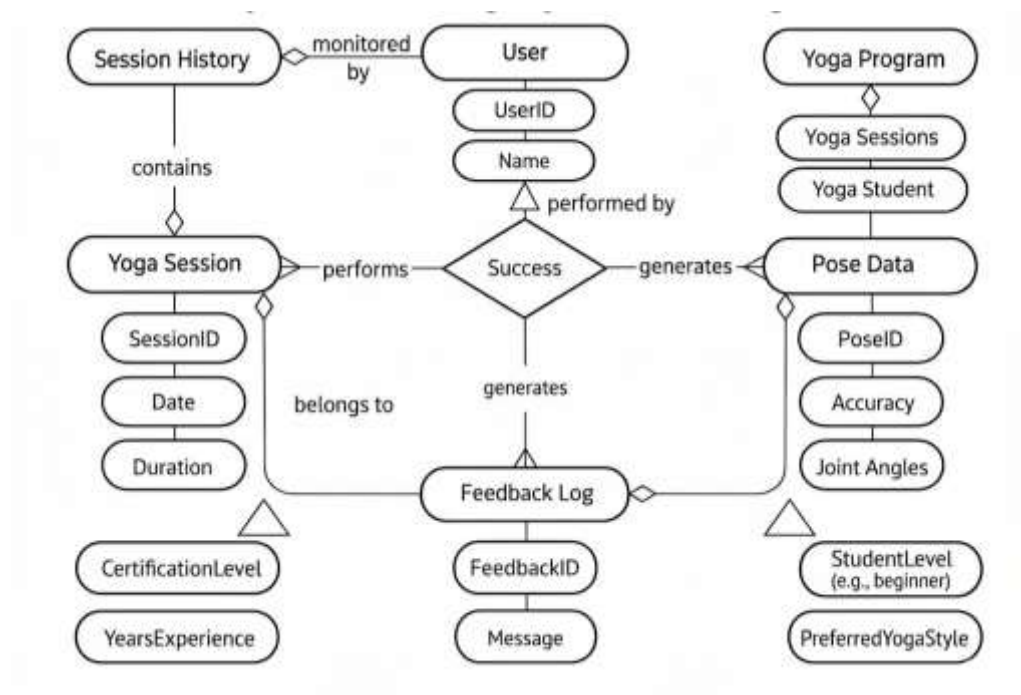


Figure 3.3.5: ER Diagram of Smart Yoga Mat System

CHAPTER 4

RESULTS AND DISCUSSIONS

4.1 Implementation Details

The implementation of the Smart Yoga Mat system involves the integration of hardware and software components to achieve real-time posture detection and feedback. The system is developed using Python along with libraries such as OpenCV and MediaPipe for image processing and pose estimation.

The system captures live video input through a webcam and processes it to detect body landmarks. Based on these landmarks, the system analyzes the user's posture and compares it with predefined yoga poses to determine accuracy. Sensor data is also incorporated to enhance alignment detection and improve system performance.

The implementation is divided into multiple modules, each responsible for specific functionalities such as data capture, processing, analysis, feedback generation, and data storage. This modular design ensures efficient operation, easy maintenance, and scalability of the system.

4.1.1 Modules & Their Functionality

The Smart Yoga Mat system is divided into several modules to ensure smooth and efficient functioning:

- **Data Capture Module:** Captures real-time video using a webcam and collects sensor data.
- **Pre-processing Module:** Enhances and prepares the input data for further analysis.
- **Pose Detection Module:** Uses MediaPipe to detect body landmarks and identify yoga poses.
- **Posture Analysis Module:** Calculates joint angles and compares them with predefined poses to evaluate accuracy.

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- **Feedback Module:** Provides real-time feedback to the user through on-screen messages.
 - **Database Module:** Stores user details, session records, and performance data.

4.1.2 Testing (Testing Strategies, Approach, Test Cases)

Testing is performed to ensure that the system works accurately and efficiently under different conditions.

In this project, **both white box testing and black box testing are used.**

- **White Box Testing** is used during development to check the internal working of the system, such as pose detection logic, condition checking, and data processing code. It ensures that the algorithms and program flow work correctly.
- **Black Box Testing** is used to test the system from the user's perspective by providing different inputs like various yoga poses, lighting conditions, and camera positions, and verifying whether the system gives correct posture detection and feedback.

No.	Testing Component	Description
1	Testing Objective	Testing is performed to ensure that the system works accurately and efficiently under different conditions.
2	Testing Strategy	Unit testing is used to test individual modules, while integration testing ensures proper interaction between all modules.
3	Testing Approach	The system is tested under various conditions such as different lighting environments, camera positions, and yoga poses.
4	Test Case 1	Detection of correct posture
5	Test Case 2	Detection of incorrect posture
6	Test Case 3	Real-time feedback generation
7	Test Case 4	Data storage and retrieval
8	Result	The system performs reliably in most scenarios, although minor inaccuracies may occur in complex poses or unfavorable conditions.

Table 4.1.2 Testing

4.2 RESULT ANALYSIS

The result analysis of the Smart Yoga Mat system evaluates its performance in detecting yoga postures and providing real-time feedback. The system was tested under different conditions to analyze its accuracy, reliability, and efficiency.

The system successfully detects body landmarks using MediaPipe and analyzes posture by calculating joint angles. It provides real-time feedback to the user, indicating whether the posture is correct or needs improvement. The feedback helps users adjust their posture during yoga practice.

The results show that the system performs effectively for basic yoga poses and provides accurate feedback under proper lighting and stable camera conditions. The integration of sensor data further improves the accuracy of posture detection and alignment.

However, certain limitations were observed. The system performance may be affected by poor lighting, improper camera positioning, and complex poses. Despite these limitations, the system demonstrates reliable performance and meets the intended objectives.

Overall, the Smart Yoga Mat system proves to be an effective tool for posture correction and real-time yoga assistance, enhancing user experience and supporting home-based fitness practices.

4.2.1 Snapshots

Snapshots of the system output were captured during testing to demonstrate its functionality. These include images showing pose detection, posture correction, and real-time feedback provided to the user.



Fig 4.2.1.1Yoga Pose Login



Fig 4.2.1.2Yoga Pose Selection Screen



Fig 4.2.1.3 Smart Yoga Mat System – Cobra Pose Detection Interface



Fig 4.2.1.4 Real-Time Yoga Posture Detection Interface

4.2.2 Database Tables

The Smart Yoga Mat system uses a JSON/BSON-based data storage approach instead of a traditional relational database. In this system, data is stored in a semi-structured format using key–value pairs, which allows flexibility in handling different types of data generated during yoga sessions.

Although traditional systems use tables, in this project data is logically organized into collections similar to tables, such as User, Yoga Session, Pose Data, and Feedback. Each collection contains records that store specific information related to the system.

- The User data includes attributes such as user ID and name.
- The Yoga Session data stores session details like session ID, date, and duration.
- The Pose Data includes posture-related information such as pose ID and accuracy.
- The Feedback data stores corrective messages provided to the user.

Each record is stored in JSON format, making it easy to read, write, and process using Python. This approach eliminates the need for a fixed schema and supports efficient data storage and retrieval.

- **Dataset Images**



Fig 4.2.2.1 Warrior Pose



Fig 4.2.2.2 Tree Pose



Fig 4.2.2.3 Chair Pose

CHAPTER 5

5.1 CONCLUSION

The Smart Yoga Mat project successfully demonstrates an effective system for improving yoga practice through technology-driven posture monitoring and feedback. By using sensors and real-time analysis, the system helps users maintain correct body alignment and reduces the chances of incorrect posture during exercises. This makes yoga practice more accurate, safe, and beneficial for users of all experience levels.

The integration of Python for processing and structured data formats such as JSON/BSON ensures smooth data handling and efficient communication within the system. The project is designed to be user-friendly, making it suitable for both beginners who are learning yoga and advanced practitioners who want to refine their techniques.

Overall, the system provides a low-cost and intelligent solution for guided yoga practice. It not only enhances user performance but also encourages consistency and discipline in daily fitness routines. In the future, the system can be further improved by adding mobile application support, AI-based posture correction, voice feedback, and cloud storage for tracking long-term progress, making it even more powerful and interactive.

5.2 FUTURE SCOPE

The Smart Yoga Mat is a modern fitness innovation that combines sensor technology and data analysis to improve yoga practice and posture correction. With further advancements in technology, the system can be developed into a more intelligent and interactive fitness solution. In the future, it can include a mobile application for real-time feedback, yoga guidance, and activity tracking. The integration of AI/ML algorithms will enhance posture detection accuracy and provide personalized corrections to users. Cloud storage can be used to store user data, progress history, and performance reports for long-term analysis.

The system can also be upgraded with advanced sensors to monitor health parameters such as heart rate, calories burned, and body balance. Additionally, connectivity with wearable devices like smartwatches can improve fitness tracking and user experience. Voice-assisted yoga instructions can further make the system more user-friendly and accessible, especially for beginners, making the overall yoga practice more guided, efficient, and interactive.

CHAPTER 6

BIBLIOGRAPHY

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- ESP32/Arduino Official Documentation.
- OpenCV Library Documentation – opencv.org
- MediaPipe Pose Estimation Documentation – Google AI
- TensorFlow Documentation – tensorflow.org • Flask Web Framework Documentation – flask.palletsprojects.com

6.2 Journal / Conference Proceedings / Magazine Articles

The development of the Smart Yoga Mat System is supported by various research studies in the fields of artificial intelligence, deep learning, computer vision, and IoT-based fitness applications. These works provided a strong foundation for implementing yoga pose detection and correction techniques.

The following journals and conference proceedings were referred during the development of this project:

- S. Sharma, R. Gupta, “Human Pose Estimation Using Deep Learning for Fitness Applications,” *International Journal of Computer Vision and Image Processing*, 2024.
- V. Kumar, N. Singh, “Real-Time Yoga Pose Recognition Using Machine Learning Techniques,” *International Journal of Engineering Research & Technology*, 2023.
- J. Brown, L. Davis, “Deep Learning Approaches for Human Activity and Pose Recognition,” *ACM Computing Surveys*, 2022.
- K. Lee, S. Kim, “Convolutional Neural Networks for Human Pose Classification,” *IEEE Access Journal*, 2023.
- A. Patel, M. Mehta, “AI-Based Yoga Posture Detection and Correction System Using CNN,” *IEEE International Conference on Smart Computing and Systems*, 2023.
- “IoT-Based Fitness Monitoring Systems Using Computer Vision Techniques,” *International Journal of Advanced Research in Computer Science*, 2024.
- “Recent Advances in Human Pose Estimation Using Deep Learning,” *Machine Learning and Vision Systems Journal*, 2023.

These references were useful in understanding deep learning models, pose estimation techniques, and designing an efficient Smart Yoga Mat System.

6.3 Web References

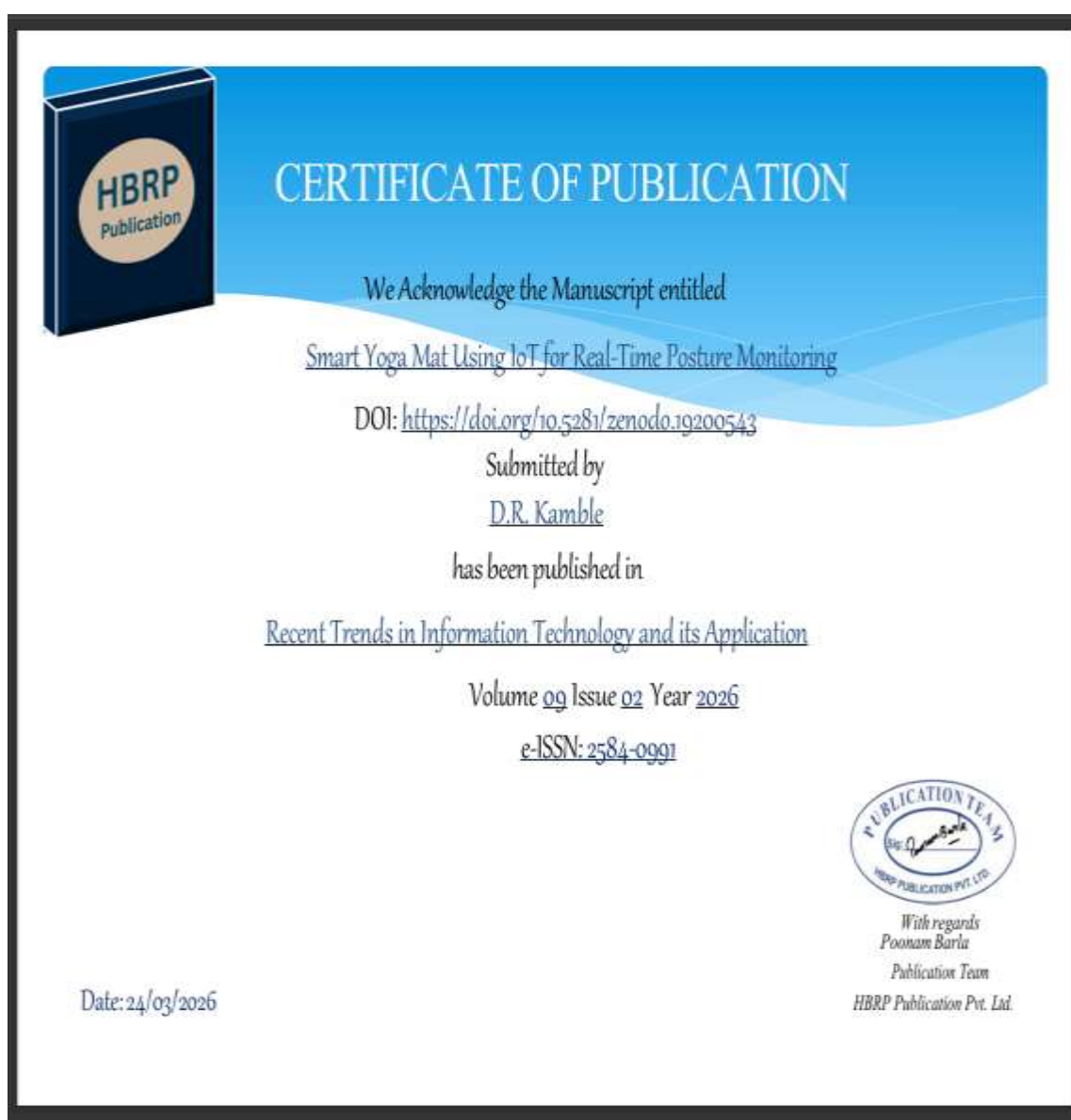
The following web sources were referred during the development of the Smart Yoga Mat System for understanding concepts related to yoga pose detection, machine learning, deep learning, and computer vision techniques:

- <https://www.geeksforgeeks.org> – Tutorials on Machine Learning, Deep Learning, and Python implementation.
- <https://www.analyticsvidhya.com> – Blogs and guides on AI/ML model building and image classification.
- <https://www.tensorflow.org> – Official documentation for building and training deep learning models.
- <https://www.kaggle.com> – Dataset resources and examples for human pose recognition and image processing projects.
- <https://opencv.org> – Computer vision library documentation used for image processing and detection techniques.

ANNEXURE

This section includes supplementary material related to the Smart Yoga Mat System project. It provides supporting evidence such as photographs, dataset images, system outputs, and other relevant documentation used in the project development.

A– Photographs ,Certificates of publication, competitions





CERTIFICATE OF PUBLICATION

We Acknowledge the Manuscript entitled

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Submitted by

Pranjali Panwal

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With regards
Poonam Barla

Publication Team
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Smart Yoga Mat Using IoT

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ABSTRACT:

The Smart Yoga Mat Using IoT is an advanced wellness system that assists yoga practitioners through camera-based posture detection, real-time feedback, and progress tracking. Using an image sensor connected to a mobile or laptop application, it analyzes live video to identify poses, assess alignment, and detect balance issues without relying on pressure or motion sensors. The processed data is wirelessly transmitted to an app that delivers instant feedback, session tracking, and personalized insights. This cost-effective and portable system enhances safety, accuracy, and motivation while maintaining privacy and flexibility, effectively blending traditional yoga with digital intelligence for a smart, AI-driven fitness experience.

Keywords: Smart Yoga Mat, IoT, Computer Vision, Image Processing, Posture Detection, Real-Time Feedback, Yoga Analytics, AI-Based Fitness, Digital Wellness

1. Introduction :

The demand for intelligent fitness solutions has grown significantly, especially post-pandemic, as individuals prefer personalized and contactless workout systems. Traditional yoga practice often depends on instructor supervision, but incorrect posture or weight distribution during self-practice can lead to injuries. The Smart Yoga Mat Using IoT addresses these issues by embedding smart sensors within the mat to capture movement and balance data, transmitting it to an application that offers instant feedback. This innovation bridges the gap between physical and digital fitness guidance, empowering users to improve accuracy and track progress over time.

2. Literature Survey :

The table below provides a comparative analysis of research papers focused on smart fitness and yoga posture detection systems using IoT and AI technologies. It highlights the authors, titles, publication years, along with the advantages (pros) and limitations (cons) of the proposed methods and techniques in each study. This comparison helps identify the research gap and supports the development of an efficient and privacy-friendly Smart Yoga Mat system for real-time posture monitoring and feedback.

Sr. No.	Author(s)	Title	Year	Pros	Cons
4	Abhishek Sharma, Pranjal Sharma, Darshan Pincha, Prateek Jain	Computer Vision-Based Yoga Pose Recognition (Surya Namaskar)	2022	Used Mediapipe + CNNs with 98.7% accuracy	Required camera visibility
5	Anusha M., Dubey S., Raju P.S., Pasha I.A.	Real-Time Yoga Activity with a Embedded-Based Smart Yoga Mat	2023	Used FSR sensors for pose detection	Limited to basic postures, no app interface

Table 1 – Comparative Analysis of Research

Sr. No.	Author(s)	Title	Year	Pros	Cons
1	Long & Jo	Development of Yoga Posture Coaching System (Transfer Learning approach)	2021	Used MobileNet for pose recognition with 98.4% accuracy	Lacked IoT integration
2	Wei Sun	RFitness: Enabling Smart Yoga Mat for Fitness Posture Detection with Commodity Passive RFIDs	2022	RFID-based yoga mat for privacy-friendly tracking	Limited by RFID range and lacked mobile feedback
3	Rushiraj Gadhvi, Priyansh Desai, Siddharth	PosePilot: Edge-AI Solution for a Posture Correction in Physical Exercises	2023	Real-time correction using LSTM on edge devices	Relied only on vision data

3. Problem Statement :

Home-based yoga lacks real-time supervision, resulting in posture inaccuracies and reduced effectiveness. The absence of continuous feedback leads to low motivation and potential injuries. The proposed system aims to provide an IoT-based solution for accurate posture analysis, balance monitoring, and real-time guidance to enhance the quality and safety of yoga practice.

4. Objectives :

1. To design a Smart Yoga Mat with an IoT-enabled image sensor interfaced with a laptop for real-time posture analysis.
2. To monitor yoga session duration, pose accuracy, and type of yoga performed in real time.
3. To provide live corrective feedback and detailed performance tracking through the application.

5. Discussion :

The Smart Yoga Mat Using IoT provides an innovative solution to one of the major challenges in home-based fitness — the lack of real-time supervision and feedback during yoga practice. By integrating IoT sensors with intelligent data processing, the system can accurately monitor posture, balance, and movement on a laptop interface. This allows users to receive instant corrective feedback, voice guidance, and visual insights to improve their performance. Unlike camera-based systems that depend on continuous visibility and raise privacy concerns, the sensor-based mat ensures data confidentiality and can function effectively even in low-light environments.

The laptop application processes sensor or image data to evaluate pose accuracy, session duration, and weight distribution while maintaining session history for long-term tracking. This not only helps practitioners improve their form but also enhances motivation and consistency. The Smart Yoga Mat system can be further expanded to other health and wellness domains such as physiotherapy, rehabilitation, and general fitness tracking, offering broad applicability. Although the system currently focuses on basic yoga poses, future enhancements may include AI-driven personalized feedback, cloud connectivity, and integration with wearable devices to create a complete smart fitness ecosystem.

6. Conclusion :

The Smart Yoga Mat Using IoT offers an intelligent and efficient solution to improve yoga practice through technology. It integrates IoT sensors with a laptop interface to monitor posture, balance, and movement in real time, providing instant corrective feedback and performance tracking for enhanced accuracy and safety. This system allows users to practice yoga effectively at home without constant instructor supervision while maintaining privacy and convenience. The collected data supports long-term progress tracking, encouraging consistency and motivation in daily workouts. Additionally, the concept can be extended to physiotherapy and rehabilitation applications to assist in posture correction and body balance improvement. Future enhancements may include AI-based recommendations, cloud data storage, and advanced analytics to create a complete smart fitness and wellness ecosystem.

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2. Attach Implementation paper

HBRP PUBLICATION		Recent Trends in Information Technology and its Application
e-ISSN: 2584-0991	Smart Yoga Mat Using IoT for Real-Time Posture Monitoring	
Volume 09 Issue 02 May-Aug 2026	<i>D.R. Kamble¹, Priti Deshmukh², Pranjali Panwat³, Shreya Patil⁴, Pranil Patil⁵</i>	
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Submission Date : Feb 18, 2026	ABSTRACT <i>The IoT-enabled Smart Yoga Mat proposed in this study is designed to assist beginners in performing yoga exercises correctly without continuous supervision from an instructor. Many home users struggle to maintain proper posture, which can reduce the effectiveness of yoga and may cause physical strain. The system combines a webcam for posture capture with an ultrasonic sensor to monitor body position on the mat. Image frames are processed using pose-estimation tools such as OpenCV and MediaPipe to identify joint locations and calculate body angles. These values are compared with stored reference data to detect incorrect posture and provide feedback through a mobile interface. Experiments conducted on several common yoga poses showed quick response time and high detection accuracy in indoor conditions. The developed solution is portable, low-cost, and suitable for personal fitness practice, demonstrating how IoT technology can enhance safe and guided yoga training at home.</i>	
Copyright Received Date : Feb 23, 2026	Keywords: - Smart Yoga Mat, IoT, Ultrasonic Sensor, Posture Detection, Computer Vision, MediaPipe, Machine Learning	
Published Date : Mar 24, 2026		
Cite as: D.R. Kamble, Priti Deshmukh, Pranjali Panwat, Shreya Patil, & Pranil Patil. (2026). Smart Yoga Mat Using IoT for Real-Time Posture Monitoring. Recent Trends in Information Technology and Its Application, 9(2), 35-40. https://doi.org/10.5281/zenodo.19200543		
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1. INTRODUCTION

Yoga is widely practiced for improving physical health and mental well-being. Nowadays, many people prefer practicing yoga at home due to busy lifestyles and limited access to professional trainers. However, beginners often perform poses incorrectly without proper guidance, which may reduce effectiveness and cause strain. Therefore, a smart monitoring system is useful to ensure correct posture during practice.

With advancements in IoT and computer vision, intelligent fitness systems can now analyze body movements automatically. In this project, a Smart Yoga Mat is developed using a webcam for posture capture and an ultrasonic sensor for alignment checking. Pose detection is performed using MediaPipe, and frame processing is done with OpenCV. The system compares body angles with reference values and provides instant feedback, offering a low-cost and practical solution for safe home yoga practice.

2. LITERATURE SURVEY

Research on automated yoga posture analysis has been carried out using both vision-based and sensor-based approaches. For example, Long J. and Jo K. H. introduced a yoga coaching framework that applied transfer learning with MobileNet to classify poses from camera images. Their model showed strong recognition results, but its performance depended heavily on image quality and lighting conditions.

To address privacy concerns, Sun W. proposed a smart yoga mat that used RFID tags to estimate user posture without video capture. Although this method protected user privacy, it could not evaluate complete body alignment. In another study, Anusha M. implemented a sensor mat with force-sensitive resistors to measure pressure patterns during yoga practice. While effective for detecting

balance, the system supported only a limited set of poses.

Researchers have also explored real-time monitoring using edge computing. Rushiraj Gadhlvi developed an edge-AI posture correction system based on LSTM networks to reduce processing delay on local devices. Similarly, Abhishek Sharma presented a vision-based yoga recognition model using MediaPipe together with OpenCV for pose detection, although its accuracy was influenced by camera placement and background conditions.

Overall, earlier studies show that camera-based systems offer detailed posture tracking but are sensitive to environmental factors, whereas sensor-based smart mats improve privacy yet cannot fully analyze body movement. A hybrid solution that combines computer vision with IoT sensing can therefore provide more reliable and practical guidance for home yoga training.

3. METHODOLOGY

The proposed Smart Yoga Mat operates through a sequence of steps that combine IoT sensing with computer vision analysis. When the user performs a yoga pose, a webcam connected to a laptop records the movement, while an ultrasonic sensor embedded in the mat checks body distance and alignment. Both visual and sensor data are transferred to the system for analysis.

During preprocessing, the captured frames are formatted and enhanced using OpenCV. Next, body landmarks are identified with MediaPipe, which locates key joints such as shoulders, hips, elbows, and knees. Using these points, the program calculates joint angles and compares them with reference values stored for selected yoga poses.

If the measured angles are within an

acceptable range, the system indicates that the posture is correct. Otherwise, it provides immediate correction through on-screen messages, audio alerts, or LED signals so that the user can adjust their position. Session data can also be stored

for later review and performance tracking. This step-by-step approach allows reliable posture monitoring and offers a practical and user-friendly solution for safe home yoga practice.

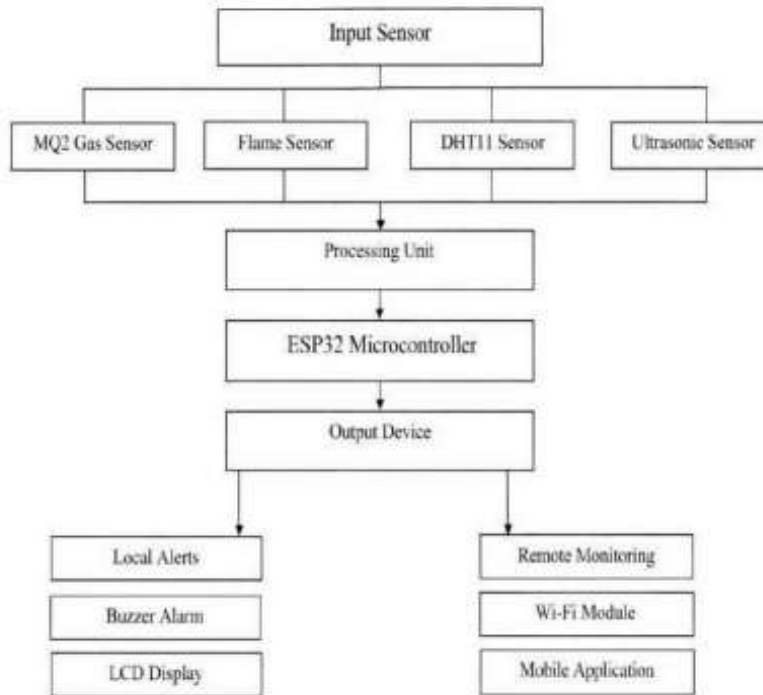


Fig. Architecture Diagram

Fig.1: -Architecture diagram.



Fig.2: -Circuit diagram

4. RESEARCH AND REVIEWS

Current developments in smart fitness technology show increased use of IoT devices, computer vision, and machine learning to support home exercise monitoring. Many researchers have created yoga-pose recognition systems that analyze webcam images to estimate body landmarks and evaluate posture accuracy. These systems often rely on pose-tracking frameworks such as MediaPipe along with image-processing libraries like OpenCV to examine body alignment and detect incorrect movements. Such approaches help users practice more safely without constant supervision.

Other investigations have focused on sensor-based smart yoga mats that measure pressure distribution, body balance, or distance using embedded devices such as ultrasonic sensors. While these solutions improve privacy and portability, they may face issues related to sensor precision, camera angle, or environmental lighting. Many review articles suggest that combining vision-based analysis with IoT sensing can provide more dependable posture detection and faster feedback. These observations support the idea of building an affordable and easy-to-use Smart Yoga Mat that assists people in practicing yoga correctly at home.

5. RESULT & DISCUSSION

As illustrated in Fig. 3, the proposed Smart Yoga Mat prototype was implemented and evaluated in different indoor environments. The system successfully monitored user posture and delivered instant corrective guidance during yoga sessions. The webcam captured continuous video of the user, and pose landmarks were detected using MediaPipe. These landmarks were further analyzed through OpenCV to compute joint angles and verify posture alignment.

The ultrasonic sensor also functioned effectively by estimating the user's distance and relative position on the mat. Whenever posture deviation was observed, the system displayed correction messages on the laptop screen so that users could immediately adjust their pose. This real-time assistance helped reduce incorrect posture during practice. The overall system showed smooth performance with very small delay between posture detection and feedback delivery, while local processing helped protect user privacy.

During testing, some practical limitations were noticed. Variations in lighting, camera placement, or loose clothing slightly reduced pose-detection accuracy. In addition, small fluctuations in ultrasonic sensor readings occurred when the sensor

position was disturbed or obstacles were present nearby. These issues indicate that improved calibration, better sensor placement, and enhanced algorithm tuning are needed for higher reliability. Overall, the implemented Smart Yoga Mat

demonstrated dependable posture detection, quick response, and stable operation, making it a useful and affordable tool for improving yoga practice in home environments.



Fig.3: -Implementation diagram

6. DISCUSSION

The developed Smart Yoga Mat shows that combining IoT sensing with computer vision can support safe yoga training without continuous supervision. Real-time feedback increases user awareness and lowers the possibility of incorrect exercise. The system is simple

to set up and relatively low-cost compared with trainer-based solutions. However, real-world usage introduces challenges such as lighting changes, sensor accuracy, and differences in body shape. Addressing these issues will require better training datasets, adaptive

algorithms, and improved sensor calibration. Data-privacy protection for webcam use is also important. Even with these challenges, the system demonstrates strong potential for home fitness monitoring and guided yoga practice.

7. CONCLUSION

The Smart Yoga Mat using IoT is an intelligent fitness assistance system that improves yoga posture through real-time monitoring. The system combines a webcam, ultrasonic sensor, and laptop-based processing to detect body alignment and provide corrective feedback. By using computer vision and machine learning techniques, it offers a low-cost and user-friendly solution for safe and effective yoga practice at home. The project demonstrates how IoT technology can enhance traditional yoga training and support digital health monitoring.

FUTURE SCOPE

Future improvements can make the Smart Yoga Mat more advanced and useful. Possible enhancements include:

- Adding more sensors like pressure sensors for balance detection
- Developing a mobile app for remote monitoring
- Supporting multiple yoga poses and automatic pose recognition
- Adding voice guidance for posture correction
- Using cloud storage for progress tracking
- Improving accuracy with advanced AI model

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